



NUTRITIONAL ASSESSMENT (HuNu3081)

CHAPTER TWO

DIETARY ASSESSMENT METHODS

Objectives

At the end of this chapter you will be able to:

- Describe the methods used for measuring food consumption at the national, household & individual level.
- Discuss Dietary Assessment methods.
- Identify steps in dietary assessment methods.
- Describe the strength and limitations of each dietary assessment method.
- Identify the factors considered for the selection of methods used to measure food consumption.

Introduction

Dietary assessments can provide information on eating habits, food & nutrient intakes of individuals. A number of tools (instruments & questionnaire) ranging from simple checklists to detailed individual weighed food records to household surveys have been developed to estimate the energy and nutrients intake.

Each tool is associated with specific strengths and weaknesses. The selected tool should be appropriate and tailored to the culture, ethnicity, age and language of the target group. For example:

- **For population groups:** interviewers of the same ethnic background, food lists & nutrients composition databases that reflect the foods consumed by the population should be used.
- **For young children:** caregivers in conjunction with the children should be used.
- **For elderly:** any special considerations (memory, special diet & supplements use) should be assessed.
- **For respondents unable to self-report:** hospital records, dietitians or nurses should be used
- **For illiterate populations:** the appropriateness of the tool used should be ensured (interviewer administered & not self-administered).

Dietary Assessment Methods include;

- ▶ Assessment of food availability at national and household level;
- ▶ Assessment of past or current intakes of nutrients from food by individuals or households in order to know food availability, consumption, and nutrient adequacy.

✚ Food consumption can be assessed at national, household, or individual level.

Measuring Food Consumption at National level

At national level Food consumption can be assessed using the following methods.

- i. **Food Balance Sheet:** also called national food accounts, food moving in to consumption, food consumption statistics & national food disappearance data.
- ii. **Total Diet Studies**
- iii. **Universal Products Code & Electronic Scanning Devices (UPCs)**

i. Food Balance Sheet

It was published by FAO since 1949. It is the most commonly used method which provides data on **the availability of food for consumption** (*i.e. food supply within a country*). It can also provide a comprehensive picture of the pattern of a country's food supply during a specified reference period but not measure the food actually ingested by population.

It is calculated from annual food production, changes in stocks, imports & exports, distribution of food within the country.

In food balance sheet, foods are grouped into 15 food groups; *i.e. cereals, roots & tubers, sugar & honey, pulses, nuts & oil seeds, vegetables, fruits, meat, eggs, fish & sea foods, milk, oils & fats, spices, stimulants and alcoholic beverages.*

ii. Total Diet Studies

Total diet studies involve:-

- ✓ Collection of large number of representative foods from a particular region or country and,
- ✓ Chemical analysis of the dietary intake of the foods.
- It can be used to **monitor** and **evaluate intake of macronutrients** & **vitamins** in population.

iii. Universal Products codes & Electronic scanning devices (UPCs)

UPCs are standard multi digit numbers with a machine readable code that represents product, size and manufacturer etc. It appear on canned & packed foods and some fresh items.

Measuring Food Consumption at the Household Level

Measuring food consumption at the Household level can be done using the following methods.

- | | |
|------------------------|------------------------------------|
| i. Food Account Method | iii. Household Food Record Method |
| ii. Telephone Survey | iv. Household 24-hrs Recall Method |

i. Food Account Method

Food account method involves daily record of all food entering in the household (purchased, received as gifts, or produced) during a given time period usually 7 days. It doesn't measure the food that is actually eaten. It is inexpensive to use for a large population.

ii. Telephone Survey

Telephone survey is used to obtain information on household food consumption or on specific foods. It is less expensive than other methods.

iii. Household Food Record Method

Household food record method involves keeping records and measuring all foods consumed at each meal time by weight, before they are divided into individual servings over at least one week period by householders or field workers.

- A 10% wastage factor of edible portion can be used to subtract food not consumed.
- This is the most accurate method of measuring household food intake (if food is weighed consistently) but puts a large burden.

$$\text{Food intake per person} = \frac{\text{total food consumed by the household}}{\text{total man value}}$$

“Man value” assigned based on sex & age: **Males>14yrs=1, females>11y and boys 11-14yrs=0.9, 7-10yrs=0.75, 4-6yrs=0.4, <4yrs=0.15.**

iv. Household 24-hrs Recall Method

In this method a household member that prepare food is interviewed about all foods consumed in the household within the last 24 hours. Taking food from **a shared bowl is a challenge**. It is used to monitor changes in household food security.

$$\text{Food intake per person} = \frac{\text{amount consumed by household}}{\text{no. of persons in household}}$$

Measuring Food Consumption of Individuals

There are three stages in a dietary assessment protocol suitable for individuals.

- Measurement of food intakes;
- Calculation of nutrient intakes; and
- Evaluation of nutrient intakes in relation to recommendations.

Methods for measuring food consumption of individuals can be either:

- ✓ Quantitative methods or;
- ✓ Qualitative methods.

The choice depending primarily on the objectives of the study.

1. Quantitative methods

These methods measure the quantity of food consumed over a one-day period and estimate prevalence of inadequate intake.

. They include:

- ▶ **24-hr recalls**:- used to assess **past intake** and;
- ▶ **Food records** (Weighed food records & Estimated food records (Food Diary)): used to assess **current intake**.

2. Qualitative methods

These methods are used most frequently to assess habitual intake of foods or specific classes of foods rather than intakes of nutrients. With some modification, however, they can provide data on habitual nutrient intakes. They provide retrospective information over longer time frame.

Qualitative methods include:

- ▶ **Food frequency questionnaire** (semi-quantitative FFQs) &
 - ▶ **Dietary history**
- ➡ Both of the methods are used to assess **past intake**.

1. Quantitative methods

1.1. Twenty-four hour recall methods

Twenty-four hour recall methods are the most widely used dietary assessment method. In these methods subjects/ their parents or caretakers (for children less than 8 years) are asked to remember their exact food intake (what they ate, drunk & what amount) during the previous 24-hr period (from sunrise time yesterday to sunrise time today).

They are quick, easy and depends on short-term memory, but may not be truly representative of the person's usual intake. To improve the recall, food models (for portion size), key questions and memory aids should be used.

It is also preferable to conduct interviews in subject's home because,

- ▶ It encourages participation,
- ▶ Improves accuracy of recall and
- ▶ Facilitates calibration of local household utensils).

▪ Important considerations while conducting 24-hr recalls

Interviewer must be:

- ✓ Trained, friendly, mature, nonjudgmental & sensitive.
- ✓ Female interviewers are preferred.
- ✓ Probe the respondent without forcing an answer on him/her.
- ✓ Food models are used to represent portion size.
- ✓ Subjects should be representative (to characterize the average usual intake of a population group).
- ✓ All days of the week should be equally represented and avoid any day of the week that may have an effect on food & nutrient intakes.

Steps of 24-hr recalls

Twenty-four hour recall methods have 4 steps.

Step 1: List all foods and drinks consumed sequentially during the previous 24-hrs starting at the time of waking.

Step 2: Describe in detail each food listed (*i.e.*, cooking methods, brand names, and time and place of consumption etc.). Recall information on ingredients of mixed dishes as well.

Step 3: Estimate portion size of each food item consumed & change in to weight equivalents.

Step 4: Review recall to check all items are recorded correctly.

- ✓ Check foods listed against picture chart; check recipes.
- ✓ Check whether recall was a “usual” day.

Table: Data sheet for 24hr records

Name:				Date:		
Street address:				Day of the week:		
Town/city:						
Place eaten	Time	Description of food or drink. Give brand name if applicable	Amount	LAB. USE ONLY		
				Day/Meal code	Food code	Amount code

Additional questions:

Was intake unusual in any way? Yes (...) No (...)

If yes, in what way?

Do you take vitamin or mineral supplements? Yes (...) No (...)

If yes, how many per day? (...) per week? (...)

If yes, what kind? (give brand if possible)

Multivitamin Iron Ascorbic acid

Other (list)

Advantages & limitations of 24-hr recall methods

<u>Advantages</u>	<u>Limitations</u>
▶ Quick (10-20 minutes) & inexpensive. ▶ Large coverage; High response rate. ▶ Low respondent burden; high response rate. ▶ Suitable for large scale surveys. ▶ Used for illiterate subjects. ▶ Doesn't alter dietary habit. ▶ Can be conducted by telephone.	▶ Relies on memory & motivation. ▶ Relies on skill of interviewer. ▶ Subject to social desirability & recall biases: may omit foods consumed (under reporting) or over estimate of dietary intake) occurs. ▶ Inaccuracies in portion size estimates, when eating from common pot & for mixed dishes. ▶ Need to convert portion sizes to weight equivalents: time consuming & prone to errors.

Different forms of 24 hr recalls

1. Multiple-pass 24-hr recall: has 3 passes.

- In the first pass: Quick list: a quick list of foods/ drinks consumed is obtained.
- In the second pass: Detailed description: detailed information about the meal/ snacks consumed (including time and place) are recorded.
- The third pass: Record review.
 - ✓ Review of all the foods eaten or drunk in chronological order.
 - ✓ Prompt for foods that may have been forgotten.
 - ✓ Prompt for any additional eating or drinking occasions or foods/drinks possibly consumed.
 - ✓ Ask the respondent to name the place where each food or drink was consumed.

2. Interactive 24-hr recall: is a modification of 24-hr recalls.

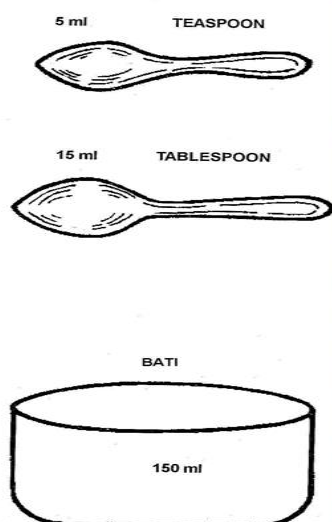
- First, you should introduce yourself and explain the purpose of recall.
- You should give respondents a picture chart of local staple foods on day before recall & instruct them to tick each food item & amounts consumed.
- You should provide bowls & plates for use on the recall days to help the respondents visualize the amount of food consumed.
- Lastly, ask respondents to serve the actual portion size of each staple food consumed into their own dishes; weigh portions.
 - ✓ Interactive 24-hr recall enhances recall of foods consumed.
 - ✓ It reduces number of memory lapses.
 - ✓ It improves portion size estimates.

3. Repeated 24-h recalls: are used to estimate usual food intake.

The number of 24-hr recalls depends on: day to day variation in food intake within one individual (*i.e.* within-subject variation). In turn affected by: nutrient under study, the study population & seasonal variations in intake.

- If possible, non-consecutive days should be selected.

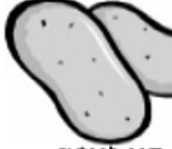
Measurement aids to quantify portion sizes in 24-hr recalls



Picture chart of local staple foods

Subject ID _____ Address _____

Mother's Name _____ Date _____ | Age of baby _____ months

 nvttech.com	$\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ whole orange		 nvttech.com	Feeding spoons of mashed pumpkin
 nvttech.com	$\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ whole banana		 nvttech.com	Feeding spoons of mashed potatoe
 nvttech.com	$\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ whole apple		 nvttech.com	Feeding spoons of boiled cabbage

1.2 Food Records

1.2.1. Weighed Food Records

In this method, the respondent or research assistant weighs all foods (including snacks), drinks consumed as well as left overs during a specified time period using dietary scales.

- Details of methods of food preparation and description of each food item including brand names will also be recorded.
- For occasional meals eaten away from home subject/ researcher records amount & description of food eaten & weighs foods consumed by the subject.

It is considered as the most precise method for assessing food intakes of individuals. **Seven-day records** is used as the “**gold standard**”.

Advantages & Limitations of Weighed Food Records

<u>Advantages</u>	<u>Limitations</u>
➤ It's more accurate & considered as gold standard. ➤ It does not rely on memory (no respondent memory loss). ➤ It can provide accurate data on portion sizes & mixed dishes. ➤ It can provide accurate data on hygiene & sanitation in home.	➤ It's Invasive; high respondent burden; ➤ It's labor intensive & expensive (costly). ➤ May change the dietary habit of respondents during the survey due to fear of burden. ➤ It need literate respondents to complete the records.

1.2.2 Observed Weighed Record Method

In this method the investigator him/herself records the amount and type of food consumed by the study subjects over specified period of time. This method is usually applied for disabled people, infants and young children, mentally ill people, institutionalized elderly people or patients admitted to a hospital.

Advantage Vs disadvantages

The same as the weighed record method.

1.2.3. Estimated Food Records (Food Diary)

The investigator (usually) ask subject(s) & record detailed description of whatever they ate/ drank for specified period of time with estimation of the portion sizes consumed. House hold measures such as cups, spoons, rulers (for meat), number (egg) are used for estimating amount of food.

While using estimated food records errors may arise:

- To quantify portion size correctly.
- Conversion of volumes to weights.
- Food diary gives relatively accurate estimate of nutrient intake if done properly.

FOOD DIARY	
Date	
Breakfast	
Lunch	
Dinner	
Snacks	

Sources of error in collecting and recording food intakes

Sources of error	24-hr recall	Estimated record	Weighed record
Omitting foods	++	+	+
Adding foods	++	-	-
Estimating food portion sizes	++	++	-
Estimating frequency of consumption	-	-	-
Day-to-day variation	++	++	++
Changes in diet	+	+	++
Coding errors	++	++	++

++ error is likely, + error is possible, - error is unlikely

2. Qualitative Methods

2.1. Food Frequency Questionnaires (FFQs)

FFQs are used to assess the frequency with which food items or food groups are consumed during a specified time period.

- ✓ Frequency of use: daily, weekly, monthly or yearly.

They provide descriptive qualitative information about usual food consumption patterns. They involve preparation of standardized self or interviewer administered food frequency questionnaires, which are based on the local staple diet to determine the frequency of consumption of a particular nutrient.

Some FFQs are designed to assess intake of individual nutrients or food components (vitamin A, fat, calcium etc...) and some are designed to assess a wide spectrum of macro & micronutrients (comprehensive assessment).

Important considerations:

- Avoid open-ended questions; preformatted lists best.
- Use well-defined foods & food categories.

2.1.1 Semi-quantitative Food Frequency Questionnaires

Sometimes the quantities consumed could be weighed and recorded, in such circumstances, the FFQs are called semi-quantitative food frequency questionnaire (SQ-FFQs).

They quantify portion sizes. Subjects are allowed to indicate whether their usual portion sizes are small, medium, large with respect to a stated standard portion for certain age/sex groups.

Advantages and limitations of FFQs

<u>Advantages</u>	<u>Limitations</u>
● They are Quick (several interviews/day) & are less costly. ● Has large coverage; low respondent burden & high response rate. ● FFQs are standardized: reduces between interviewers variations. ● Used for geographical areas with widely scattered population.	● Relies on memory: difficult for elderly, children & illiterates. ● Difficult for those eating from common pot. ● Need source of valid data for portion sizes: may overestimate nutrient intake. ● Very difficult to develop in multi-cultural society where different staple foods are consumed.

Example of Semi-quantitative FFQs for vitamin A rich foods

Food list	Frequency of consumption				Portion size consumed
	Daily	Every other day	Once per week	Once per month	
Carrot					
Cabbage					
Papaya					
Mango					
Cod liver oil					
Liver					

2.2. Dietary History

They are designed to assess/ determine the **usual nutrient intake** and **meal patterns** of an individual or a group from food over a longer period of time retrospectively (the previous month, six months, or sometimes the previous year). They are usually used to see the association between **diet** and **disease**.

- Interview method consisting 24-hr recall of actual intake, plus information on usual eating patterns (meal times & snacks) & FFQs to verify and clarify initial data on consumption of specific food items used.
- Usual portion sizes recorded in household measures and nutrient intakes calculated using food composition data.

Advantages and limitations of a Dietary History

<u>Advantages</u>	<u>Limitations</u>
▪ Provides information on habitual/ usual intakes. ▪ Relatively low respondent burden compared to food records. Used for counseling patients.	▪ Relies on respondent's memory: unsuitable for children <14 years of age and for the elderly. ▪ Time consuming: up to 2 hours. ▪ Not standardized method. ▪ Relies on household measures for portion size estimates; tends to overestimate nutrient intakes.

How to Assess Food Intakes in Young Children & the Elderly

Assessment of Dietary intake in Children

The accurate assessment of food intake in children is of concern because:-

- ✓ Dietary habits formed in early life in response to physiological requirements
- ✓ Psycho-social pressures may have considerable impact on long-term health status.

In young children, dietary intake is highly dependent on the caregiver who determines the type, quantity and quality of foods the child receives.

- Until children are aware of their food intake (approximately 7–8 years) parents' help is necessary.
- Children aged 10–12 years can report their food intake.
- Currently dietary survey methods, (constructed for adult populations), are also appropriate for collecting data from pediatric populations, (have methodological challenges).
- In children 10 years or older, **24-hour recalls** can be used.
- **Food frequency questionnaires** can be used to assess dietary intake for children older than 12 years.

Assessment of Dietary intake in the Elderly

Food intake measurement methods that rely on retrospective reports, may be less suitable in older people. In healthy & well-functioning elderly, well-conducted simple methods (**24 hr recall** & **FFQs**) may give more accurate information.

- **Dietary histories** seem to **underestimate food intake in the elderly**.

N.B. None of the dietary assessment methods give accurate estimates of the usual energy requirements of individual elderly subjects.

Selecting an Appropriate Dietary Assessment Method

Before implementing a dietary assessment tool, consider:

- Whether or not you are interested to assess habitual intakes; or
- Subject's routine eating patterns; or
- Capturing exactly what they consumed immediately prior to a study visit?
- How many days of intake are necessary?
- The literacy of population under study & their characteristics (such as age).

The number of measurement days, their selection & spacing, and the method chosen for measuring food consumption of individuals depends on a variety of factors; including:

- Level of study objectives.
- Validity and reproducibility of the method.
- Measurement errors: both systematic & random errors.
- Characteristics of the subjects within the study population.
- Respondent burden of the method & available resources.

Generally, the more accurate methods are associated with higher costs, greater respondent burden, and lower response rates.

Major sources of error in the collection and recording of food composition data

- **Non-response bias:** lack of response or poor compliance (results in non-representative sample).
- **Respondent bias:** leads to over/ under estimate of facts (income & age), overestimate consumption of good foods and under report consumption of bad foods, alcohol & etc.
- **Interviewer bias:** occurs w/n different interviewers probe information to varying degree, intentionally omit certain questions or recording responses incorrectly.
- **Respondent memory lapses:** unintentional omission or addition of foods; for recalls, FFQs, Dietary History only.
- **Omission of supplements:** cause significant error in calculating nutrient intakes.
- **Incorrect estimation of portion sizes & converting portion sizes to weight equivalents:** for recalls, FFQs, Dietary History only.
- **Errors in handling of mixed dishes.**
- **The ‘flat slope syndrome’:** a tendency to overestimate low intakes & under estimate high intakes in recall methods.
- **Coding and computation errors:** arise when portion size estimates are converted from household measures into grams or food items are incorrectly coded (2% milk coded as whole milk).

Strategies to avoid/ minimize measurement errors in Dietary Assessment

- **Non-response bias:**
 - ✓ Minimize respondent burden, mail or telephone follow-ups, offering material rewards; incentives, train interviewers.
- **Respondent and interviewer biases:**
 - ✓ Training interviewers, conduct private interviews, developing standardized interviewing techniques & questionnaire, avoid value judgments etc.

- **Respondent memory lapses:**
 - ✓ Probing questions using memory aids, use visual aids (food models or photographs), multiple-pass recall interviews etc.
- **Incorrect estimation of portion sizes:**
 - ✓ Use of graduated food models & measure household measures.

Reproducibility of Dietary Assessment methods

- Diet varies dramatically from day to day.
- Identical interview situations are never achieved in practice.
- It is difficult to **distinguish real changes** in diet from errors in measurement.
- Repeat interviews should ideally be so close as to influence answers.

Validity of Dietary Assessment methods

Validity: the degree to which a dietary assessment method measures what it intends to measure.

Validation studies of dietary records are difficult and few have been done. There is no ideal method of dietary assessment against which all other methods may be calibrated.

Criteria and Rationale to Evaluate Interviewer Skills

- **Privacy of interview:** to avoid disturbances thus hindering recall.
- **Introduction & manner of interviewer:** enhances data completeness & accuracy.
- **Manner of questioning & use of non-directed questions:** avoids bias; for example:
 - Did you add anything to your toast? but NOT Did you add butter on your toast?
 - How much of that did you eat? but NOT Did you eat all your toast?
- **Pace:** too fast – guesswork; too slow – boredom.

Calculation of Nutrient Intakes

Nutrient intakes can be calculated from food consumption data. Food composition values, representative of the average composition of a particular foodstuff on a year-round nation-wide basis. These values generally indicate the total amount of the constituent in the food, rather than the amount actually absorbed. Hence, intakes of most nutrients calculated from food intake data, represent the maximum available to the body.

- Computer software program used to calculate nutrient intakes: **NutriSurvey (NS), Food processor, ENA for SMART** & a nutrient database.

Before selecting a computer program, consider:

- ✓ The source of the accompanying nutrient database, its completeness (in terms of foods listed & availability of the nutrient values for individual foods), its validity for calculating the nutrient intakes.

There are two stages in the computer calculation of nutrient intakes:

1. Coding the food intake data into a defined machine readable form; and
2. Using a software package to calculate energy and nutrient intakes from the food intake data.

Generally, coding is done manually and usually involves assigning a numerical code to the subject, day, meal, food, and amount. The food identification code can define the food group major and minor subgroup, and finally the individual food. The code may also provide information on preserving or processing techniques, storage conditions, etc.

Evaluation of Nutrient Intakes from Food Consumption Data

This involves comparison with tables of recommended nutrient intakes. Recommended nutrient intakes are set for a particular group of healthy individuals with specified characteristics, consuming a typical dietary pattern of the country. When tables of recommended intakes for a particular country are not available, the FAO and/or WHO requirements are often used. All the methods used to evaluate nutrient intakes provide an estimate of the risk for nutrient inadequacy.

Steps in Assessment of Nutrient Intakes

1. Measuring food intake.
2. Converting foods to nutrients; conversion of portion sizes to weight equivalents and coding the data and calculating nutrient intakes.
3. Assessing distribution of “usual” nutrient intakes.
4. Evaluating dietary adequacy.

Sources of Food Composition Values in Tables or Data Bases

- Based only on ‘in-house’ chemical analysis of locally grown foods.
 - ✓ Ethiopian, Zambian & Malaysian Food Composition Table, UK McCance & Widdowson’s the Composition of Foods.
- Compiled from in-house chemical analyses plus other sources (industry; government labs; contract research; published literature; other Food Composition Tables (FCTs).
 - ✓ USDA nutrient database for standard reference, New Zealand Food Composition Tables and Thai FCT, Indonesian FCT.
- Values taken exclusively from other tables.
 - ✓ Cuba

End of Chapter Two